

# AsanaAI: A Privacy-Preserving, Occlusion-Aware and Safety-Screened Yoga Posture Coach for Commodity Hardware

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**Abstract**—Self-guided yoga can cause injury, and the camera-based coaches published in 2025–2026 share open problems: accuracy is reported on random splits of small image sets, joints the camera cannot see are still judged, one body model serves everyone, generated advice is not screened, and systems either upload video or need extra hardware. We present AsanaAI, a yoga posture coach that runs in a web browser on commodity phones and laptops and sends only 33 body-landmark coordinates, never images, to a small CPU server. It combines supervision from instructor videos verified by three independent signals (speech, biomechanical rules, steadiness), a name-and-veto cascade of two 0.36 M-parameter residual MLPs that can abstain, a 0.89 M-parameter skeleton-graph network that separates held poses from transitions, visibility-aware scoring that declines to judge hidden joints, a personal range-of-motion profile, and a safety-screened English/Hindi/Bengali guidance pipeline. We evaluate only on data the models never trained on. The cascade names the pose on 78.9% of 1,685 held-out photos with a 20.8% false-alarm rate on other poses and seven poses at recall and precision  $\geq 0.70$ , against 36.9%, 78.3% and none for the rule engine alone; a random forest on the same data is stronger on photos from the training sources (85.3%) but weakest on wild photos from another source (35.0% against 55.3%), so we report both. On held-out videos the skeleton-graph network reaches 78.0% macro recall, but under one common recipe simpler sequence models, including an 86 k-parameter MLP on window statistics, match or exceed it on held poses and on a rule-defined movement task, and a time-reversal test is at chance for every model, so we claim neither that the graph nor that frame order is necessary; what the sequence input does contribute is depth, since removing the  $z$  coordinate costs every sequence model 4 to 60 points of macro recall; our labels name only one transition, so movement modelling is future work. Server compute is 7–9 ms per frame within 324 MB, a mid-range phone extracts landmarks at 19–24 results/s, and eight of eleven poses pass an end-to-end photo test on the phone. We state the limitations: no live-person study, weak mountain, cobra and child’s pose, and a form score that is blind to arm errors.

**Index Terms**—yoga posture correction, pose estimation, skeleton

graph network, occlusion, personalisation, safe feedback, on-device inference

## I. INTRODUCTION

Yoga is practised worldwide and is increasingly learned alone, from videos and phone apps; a systematic review of published case reports documented adverse events associated with yoga [1]. A camera-based coach that tells a practitioner which joint is off, in their own language, is therefore attractive, and real-time body-landmark estimators that run on a phone [2], [3] make it technically possible. A wave of 2025–2026 systems has followed: sensor-augmented and IoT designs [4]–[7], a virtual-reality coach [8], edge and hybrid pipelines [9]–[11], joint-level correction from a single image [12], an accessibility-oriented system [13], and a clinical validation of on-device pose estimation for resistance training [14].

*Open problems.* Reading this literature, we identify six problems that remain open (Table I summarises what each system reports).

- (G1) Evaluation. Reviews note heterogeneous metrics and uncertain generalisability [15], [16], and typical results are measured on a random split of a small image set (for example 96.09% on a 256-image test split of 1,280 images [17]); random splits place near-identical frames and photos on both sides and are a known source of inflated results [18]. None of the abstracts we read reports how often a pose outside the vocabulary is wrongly named.
- (G2) Temporal behaviour. Pose transition is described as under-explored [19]; recurrent models appear in several systems [9], [10], but their abstracts report no evaluation on videos of unseen people.
- (G3) Personalisation. Where recent yoga systems personalise, they rely on extra sensors [7], [11].

- (G4) Hidden joints. Occlusion is a stated difficulty of the domain [20], yet the yoga systems above do not describe in their abstracts how a hidden joint is scored.
- (G5) Safety of generated advice. Vision–language coaching still shows substantial gaps against human coaches [21], and a system that uses a language-model voice assistant [8] reports no screening of its output in its abstract.
- (G6) Deployment. Systems need mats, headsets, robots or wearables [5]–[8], and none of the abstracts we read reports guidance beyond English.

*Approach and research questions.* We propose AsanaAI, a coach designed around these problems and evaluated only on data its models never trained on. We ask:

- RQ1: Does a name-and-veto cascade of small residual MLPs name poses accurately on held-out photos while rarely naming a pose that is not in its vocabulary, compared with standard classifiers?
- RQ2: Does a skeleton-graph temporal network recognise held poses and separate them from transitions on videos of unseen people, what input rate does it need, and do the skeleton graph or the frame order add accuracy over simpler models, on held poses and on movement?
- RQ3: Is declining to score hidden joints preferable to recovering them by mirroring or by mesh recovery?
- RQ4: Does a personal range-of-motion profile change feedback only where intended, and can the system run on commodity hardware with acceptable latency?

*Contributions.*

- 1) *Cue-verified supervision:* instructor videos are turned into training labels only where a speech transcript, a biomechanical rule engine and a steadiness test all agree.
- 2) *A name-and-veto cascade* of two 3-head residual MLPs with an explicit “not one of mine” answer, and a controlled comparison with five standard classifiers on two held-out photo sets, including a result that goes against our own design (Section V-A).
- 3) *A skeleton-graph flow check* that separates held poses from transitions, with its input-rate requirement measured and a controlled comparison with four simpler sequence models and a graph-free variant on held poses and on a rule-defined movement task, which shows parity on held poses and no advantage on movement.
- 4) *Occlusion-aware scoring* that abstains on hidden joints, compared with mirroring and mesh recovery.
- 5) *Personal calibration and safety-screened guidance* in English, Hindi and Bengali, with the image never leaving the device.
- 6) *Evidence on commodity hardware:* server and phone measurements and an end-to-end test on a mid-range phone, with live screenshots of the deployed interface.

## II. RELATED WORK

*Yoga recognition and datasets.* Yoga-82 provides 82 poses with a three-level hierarchy [20]; 3DYoga90 adds 90 poses as

RGB video and 3-D skeleton sequences [22]; recent classifiers report 96.09% on a 16-pose image set [17] and 90.7% top-1 on Yoga-82 with joint-level correction from a single image [12]. These are static, single-frame results on random or standard splits, with no measure of false alarms on poses outside the vocabulary.

*Posture-feedback systems.* Table I compares recent systems using only what their abstracts report. Sensor-based designs use IMUs [5], ultrasonic sensors [6] or pressure mats [7]; YogaVR adds a VR headset and a camera robot with a language-model voice assistant [8]. Vision pipelines use recurrent temporal models [9], [10], graph and temporal convolutions for sports posture [4], or a vision–IMU graph network for personalised posture identification [11]; one system targets users with lower-body disabilities [13]. For resistance training, on-device pose estimation has been validated clinically [14], and a vision–language model coach shows substantial gaps against human coaches [21]. A benchmark of correct and incorrect isometric poses provides labelled form data [23].

*Temporal modelling and occlusion.* ST-GCN applies graph convolution over a skeleton and temporal convolution over time [24]; pose transition in yoga has been posed as a generation task [19]. Mesh-recovery models fit a body model [25] and, like CLIFF [26], can place hidden joints plausibly but need the image; we measure CLIFF against a trivial left–right mirror (Section V-C).

*Evaluation practice.* Leakage inflates results across fields [18]; selective classification gives a classifier the option to abstain [27], which is the role of the “not one of mine” veto in our cascade.

## III. PROPOSED APPROACH

### A. Overview

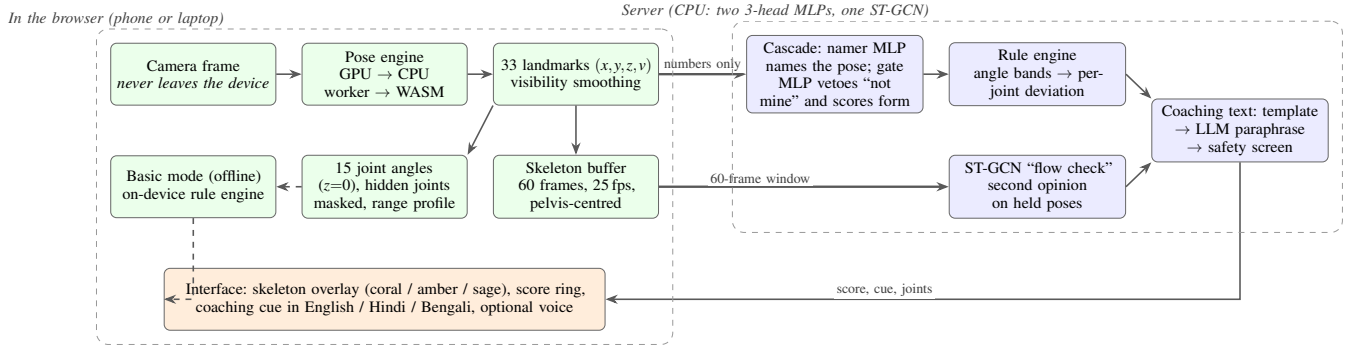
Fig. 1 shows the data flow. The browser runs a pose engine and obtains 33 landmarks  $(x, y, z, v)$  per frame, with visibility  $v$  smoothed by hysteresis so that a joint does not flicker between seen and hidden. From them it computes 15 joint angles (elbow, shoulder, hip, knee and ankle on each side, two trunk angles, the neck angle and two hip-abduction angles, from the 2-D coordinates) and a rolling 60-frame skeleton buffer resampled to 25 fps. Only these numbers leave the device; the camera frame is never uploaded. The server answers with a pose name or an explicit “not one of mine”, a form score, per-joint deviations, an optional flow-check answer and a coaching sentence. Three design rules follow from the open problems of Section I: abstain rather than guess (G1, G4), keep models small enough for a CPU (G6), and screen everything generated (G5).

### B. Cue-verified supervision

Training labels come from 24 instructor-led videos (1,206,391 frames). A frame is a positive example of a pose only if (i) the instructor names the pose in an automatic transcript [28] (English and Hindi), (ii) the rule engine of Section III-D agrees with the name, and (iii) the body is held steady. About 80 k frames qualify; about 421 k are labelled transition and

**TABLE I:** What recent systems report (from their abstracts or full text as cited; n/r = not reported in the sources we read). The last row is this work.

| System                  | Sensing and hardware                                               | Temporal modelling                     | Personalisation               | Hidden joints                    | Safety of generated advice                     | Reported evaluation                                              |
|-------------------------|--------------------------------------------------------------------|----------------------------------------|-------------------------------|----------------------------------|------------------------------------------------|------------------------------------------------------------------|
| GTA-Net [4]             | camera + IoT, 3-D pose; sports, not yoga                           | GCN + TCN                              | n/r                           | robustness to occlusion reported | n/a                                            | MPJPE on Human3.6M, HumanEva-I, MPI-INF-3DHP                     |
| Posture Guard [5]       | IMU wearable (MPU6050, ESP8266)                                    | immobility timing                      | n/r                           | n/a                              | n/r                                            | ~90% posture detection accuracy                                  |
| Smart Yoga Mat [6]      | webcam + ultrasonic sensor; MediaPipe angles vs. stored references | n/r                                    | n/r                           | n/r                              | n/r                                            | “high detection accuracy” (no figure)                            |
| Smart Yoga Mat (AI) [7] | pressure, humidity, temperature sensors; MLP                       | n/r                                    | session adapted to user state | n/a                              | n/r                                            | n/r                                                              |
| YogaVR [8]              | VR headset, PoseNet on a camera robot                              | n/r                                    | recommendations               | multi-angle camera               | n/r                                            | acceptance 8.1/10 ( $n=30$ )                                     |
| PosePilot [9]           | edge devices; LSTM, BiLSTM + attention                             | recurrent                              | n/r                           | n/r                              | n/r                                            | n/r                                                              |
| Halдар [10]             | edge AI; CNN-LSTM; quantisation                                    | recurrent                              | n/r                           | n/r                              | n/r                                            | no figures in abstract                                           |
| Vallabhaneni [11]       | vision + IMU; GCN, LSTM, attention                                 | LSTM on IMU                            | user-specific features        | n/r                              | n/r                                            | accuracy and false-alarm rate (abstract)                         |
| SAGE-Yoga [12]          | single RGB image                                                   | none                                   | n/r                           | n/r                              | n/r                                            | 90.7% top-1 on Yoga-82                                           |
| Aggarwal [13]           | MoveNet Thunder; chair yoga                                        | n/r                                    | lower-body disability         | n/r                              | n/r                                            | n/r                                                              |
| FormCoach [21]          | vision-language model; 22 exercises                                | n/r                                    | personalised corrections      | n/r                              | substantial gaps vs. humans                    | rubric-based benchmark                                           |
| AsanaAI (this work)     | browser landmarks only, CPU server; commodity phone or laptop      | skeleton-graph flow check on 60 frames | personal range profile        | abstains on hidden joints        | term-screened templates, English/Hindi/Bengali | held-out photos and videos, false-alarm rate, standard baselines |



**Fig. 1:** AsanaAI architecture. The browser turns the camera frame into 33 landmarks with an automatically chosen pose engine, masks hidden joints and sends only numbers. The server holds two small MLPs, a skeleton-graph network and a rule engine; a rule-engine port in the browser keeps a basic mode working offline (dashed).

about 705 k are excluded as ambiguous. No single source is trusted alone: an instructor may name a pose before or after the body reaches it, and labels from a rule engine alone would teach a model only to copy the rules. Training angles are computed exactly as at serving time (with  $z$  set to zero): angles from raw 3-D coordinates differ from the served ones by a median of  $14^\circ$ , which would create a train/serve gap.

### C. Name-and-veto cascade

The pose model maps the 15 angles through a 256-unit input layer (Linear, batch normalisation [29], GELU [30]) and two residual blocks [31] to three heads (Fig. 2a): a pose head ( $P$  logits), a form head (a correctness probability) and a deviation head (15 per-joint deviations). Two models of this architecture form a cascade. The *namer*, trained on photos, names the pose. The *gate*, trained on the cue-verified video table plus training photos and photos of other poses (labelled transition/unknown), answers “is this one of my poses at all?” and supplies the form score. The rule is fixed in advance: the answer is the namer’s top pose unless the gate’s top class is transition/unknown, in

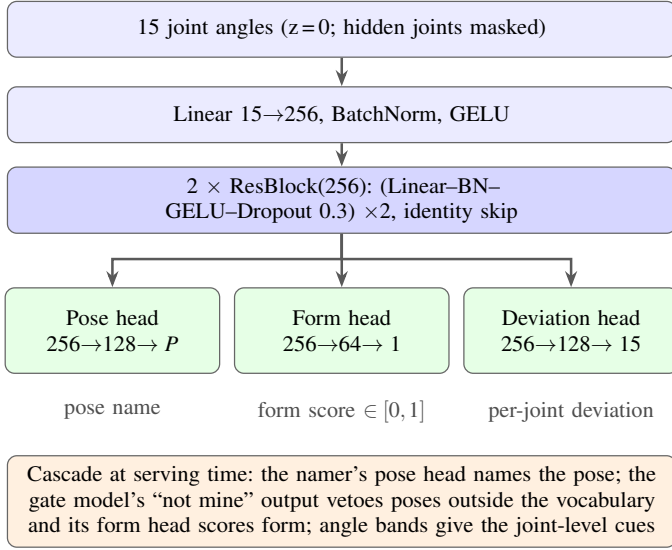
which case the system declares that it does not recognise a pose. This is a reject option in the sense of selective classification [27] applied at the system level; it was chosen on a validation half of the held-out photos and confirmed on the untouched half (Section V-A).

### D. Joint-level feedback from angle bands

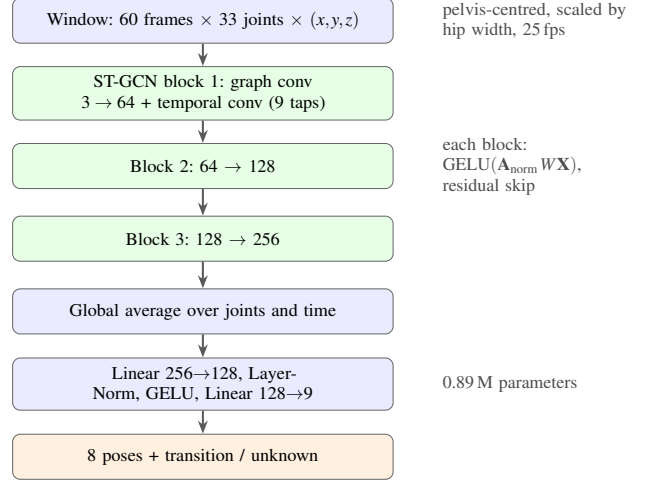
Each of 17 poses has a table of angle bands (for Downward Dog: knee  $[110^\circ, 180^\circ]$ , shoulder  $[95^\circ, 180^\circ]$ , hip  $[20^\circ, 140^\circ]$ ). A joint’s deviation is its distance in degrees outside its band, and the overlay colours each limb coral above  $15^\circ$ , amber above  $7^\circ$  and sage otherwise. The cue names the joint with the largest deviation above  $10^\circ$ . The learned deviation head is not used for feedback because it is close to chance (Section V-D); Tree and Lunge, which have no band table, therefore receive a pose-level score and neutral limbs but no per-joint cue.

### E. Skeleton-graph flow check

A 60-frame window (pelvis-centred, divided by hip width) is processed by three ST-GCN blocks [24], each a graph



(a) The 3-head residual MLP and the cascade.



(b) The skeleton-graph flow check.

**Fig. 2:** The two learned models.  $P$  is the number of pose classes. For a vocabulary of 20–25 classes the MLP has about 0.36 M parameters, and the nine-class ST-GCN about 0.89 M (counted from the model definitions).

convolution with the symmetric-normalised skeleton adjacency  $\mathbf{A}_{\text{norm}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$  followed by a 9-tap temporal convolution, GELU and a residual connection (channels  $3 \rightarrow 64 \rightarrow 128 \rightarrow 256$ ), global average pooling and a two-layer head that outputs eight target poses plus transition/unknown (Fig. 2b). The answer is shown only above a confidence of 0.70 (0.55 for child’s pose); otherwise the interface reports a static check and the per-frame cascade alone names the pose and scores form. The model is trained on 60 consecutive native-rate frames ( $\approx 2.4$  s at 25 fps), so the browser fills a time-based buffer from every camera frame and resamples it to 25 fps; Section V-B measures why this matters.

#### F. Occlusion-aware scoring

Joints whose smoothed visibility is low are neither scored nor coached; the interface lists what it is not checking and draws hidden limbs faintly. The map from each hidden joint to the angle features it affects was verified against the reference angle code by perturbation (no visibility errors). Pose naming can still be influenced by a hallucinated hidden joint; only scoring and coaching are masked. The alternatives, recovering hidden joints by mirroring or by mesh recovery, are compared in Section V-C.

#### G. Personal range-of-motion profile

At camera start the user may stand comfortably for 15 s while the 15 angles are recorded. For each joint the profile stores the observed minimum and maximum widened by  $15^\circ$  (clamped to  $[0^\circ, 180^\circ]$ ) and the mean as the resting angle; it is discarded if fewer than 20 frames were captured. The profile stays in the browser’s local storage and accompanies each request. On the server a joint whose angle lies inside its personal range has its deviation set to zero, and a personal form score closes the fraction of the gap to a perfect score that the forgiven deviation

represents, so it is never lower than the universal score and equal to it when nothing is forgiven.

#### H. Safety-screened multilingual guidance

The cue passes three stages. (1) A symbolic stage selects the joint and a reviewed template in English, Hindi or Bengali. (2) Optionally a hosted language model (a 27-billion-parameter Qwen model, temperature 0.2) may only rewrite the template. (3) A post-generation screen discards the rewrite and keeps the template if it contains a forbidden term (for example “push”, “force”, “pain”, “stretch more”, “further”). A cue that is delivered without a measured improvement escalates: it first adds how many degrees off the user is, and from the third delivery it backs off (“ease out of the shape and reset; come back only as far as feels comfortable today”). Speech uses the browser’s speech synthesis.

#### I. Implementation

The front end is a progressive web app; the back end is a FastAPI service on a CPU host. The pose engine is chosen automatically: MediaPipe on the GPU, then MediaPipe Tasks on the CPU in a pool of Web Workers (two instances on phones with six or more cores), then TensorFlow.js with WebAssembly [32]. A TypeScript port of the rule engine and templates answers locally when the server is unreachable (“basic mode”); it matches the Python original on 10,000 generated frame cases and 2,500 coaching-text cases with no mismatch.

### IV. EXPERIMENTAL SETUP

**Data.** The cue-verified video corpus of Section III supplies the training frames and the temporal windows (60 frames, stride 12; 8,002 windows; three folds split by video, so a model is tested on people and rooms it never saw). Photos supply variety: 6,616 photos from two public datasets (2,634

labelled with our poses and 3,982 of other poses, used as negatives), of which 4,931 are used for training and 1,685 are held out (1,037 of them other poses); and 422 freely licensed Wikimedia Commons photos with a frozen test split of 103, scored out-of-fold by models trained without the photo’s fold. The held-out public photos share their sources with the training photos; the Commons photos come from another source and are called *wild*. The rule engine covers 19 poses; the app offers eight guided target poses (Warrior II, Tree, Triangle, Chair, Downward Dog, Plank, Seated Easy, Corpse), and the temporal network recognises these eight plus transition/unknown.

*Protocol and metrics.* A number counts only if it was measured on data the model never trained on. A pose *passes* when its recall and precision are both at least 0.70 on at least 10 held-out photos (5 on the 103-photo set; 20 windows for the temporal network). A *false alarm* is a photo of a pose that is not in the vocabulary that the system names as one of its poses (lower is better). We also report overall accuracy and macro recall. Validation accuracy printed by a trainer on a random frame split is never used as a result.

*Baselines.* Five standard classifiers are trained on exactly the rows the gate model was trained on (136,506 rows of 15 angles) and scored with the same code:  $k$ -NN ( $k=15$ , distance-weighted), an RBF SVM, logistic regression, a 300-tree random forest and a plain MLP (256–256–128, ReLU, no residual blocks or batch normalisation). The rule engine alone is the non-learned baseline. Every run uses one random seed. For the temporal network we compare with an LSTM, a BiLSTM with multi-head attention, a temporal CNN, an MLP on the mean and standard deviation of the window, and an ST-GCN with the skeleton graph removed (identity adjacency), all on the same three video folds with one recipe (AdamW [33], cosine schedule, class-weighted loss, 10 epochs, at most 6,000 training windows per class, last epoch scored); the movement task of Section V-B uses the same recipe with 12 epochs and at most 3,000 windows per class, and a single seed throughout.

*Hardware and measurement.* Server compute is timed in-process on the 2-vCPU CPU host; request latency is the round trip of 32 sequential requests from a residential connection in India; phone throughput is measured on a POCO M7 Pro 5G (MediaTek Dimensity 7025, Android 16, Chrome 154) over a USB debugging connection with the app running in its real configuration. For the live interface a credited Wikimedia photograph stands in for the camera, so no screenshot shows a live person.

## V. RESULTS

All numbers were measured on data the models never trained on (Section IV).

### A. RQ1: naming poses on held-out photos

Table II and Fig. 3a compare the cascade, its components and five standard classifiers on the 1,685 held-out public photos and on the wild set. The cascade names the pose on 78.9% of the photos with a 20.8% false-alarm rate and seven poses at recall and precision  $\geq 0.70$  (chair, corpse, downward dog, tree,

**TABLE II:** Frame-level naming on two held-out sets (bold: best in column). Public: 1,685 photos from the training sources, 1,037 of other poses. Wild: 103 Commons photos from another source, out-of-fold. <sup>†</sup>single earlier run.

| Method                           | Overall      | Public held-out |            | Wild Overall |
|----------------------------------|--------------|-----------------|------------|--------------|
|                                  |              | False alarm     | Poses pass |              |
| Rule engine only                 | 36.9%        | 78.3%           | 0          | 37.9%        |
| Logistic regression <sup>†</sup> | 29.7%        | 95.8%           | 1          | –            |
| $k$ -NN                          | 58.6%        | 55.7%           | 1          | 50.5%        |
| SVM (RBF)                        | 58.4%        | 56.2%           | 3          | 52.4%        |
| Plain MLP                        | 80.9%        | 19.1%           | 3          | 47.6%        |
| Random forest                    | <b>85.3%</b> | <b>9.8%</b>     | <b>7</b>   | 35.0%        |
| Namer alone                      | 42.9%        | 81.5%           | 1          | <b>60.2%</b> |
| Gate model alone                 | 79.6%        | 22.6%           | 3          | 49.5%        |
| Average of namer and gate        | 65.2%        | 47.7%           | 5          | –            |
| Veto lifted when rules agree     | 65.9%        | 42.3%           | 5          | –            |
| Cascade (ours)                   | 78.9%        | 20.8%           | <b>7</b>   | 55.3%        |

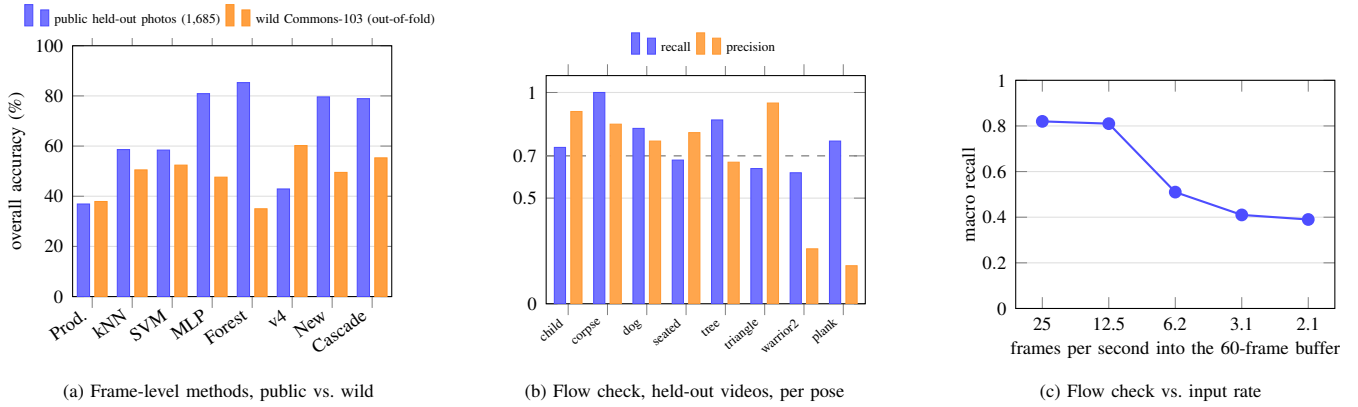
triangle, upward dog, Warrior II), against 36.9%, 78.3% and none for the rule engine alone. The veto is what matters: the namer alone recalls poses well but accepts nearly everything (81.5% false alarms), and the gate model alone is accurate but passes fewer poses (three). The decision rule was chosen on a validation half of the photos (79.3%, 20.1%, three poses) and held on the untouched half (78.6%, 21.6%, four poses).

*A result against our design.* A random forest on the same 15 angles beats the cascade on photos from the training sources (85.3% against 78.9%, 9.8% against 20.8% false alarms). We checked whether this is memorisation of near-duplicates: only 0.3% of the held-out photos lie within 0.05 (standardised angle units) of a training photo, and the forest’s lead holds in every quartile of distance to the nearest training photo, including the farthest (84.6% against 77.0%). But on the wild set the ranking reverses: the forest is the weakest learned method (35.0%), 13–25 points below every other learned method, and the cascade (55.3%) is within five points of the best (60.2%). A plain MLP matches the residual gate on public photos (80.9% against 79.6%), so the residual trunk is not what buys accuracy; what the 3-head design adds is the form and deviation outputs from one small forward pass. We conclude that no single classifier is best on both sources, that the cascade is within seven points of the best on each and keeps false alarms near 21%, and that a tree–network ensemble is the obvious next experiment.

*Why random splits mislead.* A model of the same architecture trained and validated on a random split, as is common, reached 91.5% validation accuracy but named only 26.8% of 422 unseen photos correctly, and an ST-GCN validated the same way reached 82.6% validation accuracy but only 11.6% macro recall on held-out videos.

### B. RQ2: the skeleton-graph temporal network

On three folds split by video, the network reaches 82.2% overall; because 60% of the 8,002 windows are transition/unknown this flatters it, so we also report 78.0% macro recall over the nine classes (76.9% over the eight poses). Three poses clear the bar: child’s pose (recall/precision .74/.91), corpse (1.00/.85) and downward dog (.83/.77); seated easy, tree and triangle are within about 0.07 of it (Fig. 3b). The network is conservative: its misses go to transition/unknown, not to another pose. Warrior II and plank fail on precision (.26,



**Fig. 3:** (a) Overall accuracy of the methods of Table II on the two held-out sets (Prod.: rule engine only; New: gate model alone). (b) Per-pose recall and precision of the temporal network on videos it never trained on (dashed: 0.70 bar; Warrior II and plank have few windows). (c) Macro recall when the 60-frame buffer is filled at different rates.

.18): 158 transition windows are called Warrior II, and plank has only 30 windows. A less conservative threshold chosen without leakage (an offset on the transition logit, picked on two folds and applied to the third) raises the passing poses to four with 81.9% overall; no offset reaches six. At the production confidence the network answers on 60.8% of true-pose windows with 86.8% precision.

*Input rate.* Re-scoring the same videos while subsampling every  $k$ -th frame shows that the network needs windows at the rate it was trained on: macro recall is 0.82 at 25 fps and 0.81 at 12.5 fps, then falls to 0.51, 0.41 and 0.39 at 6.2, 3.1 and 2.1 fps (Fig. 3c). Feeding it one frame per server round trip would therefore silently destroy its accuracy, which is why the browser buffers every camera frame and resamples to 25 fps. *Is the skeleton graph needed?* Table IV trains six sequence models under one recipe and scores them on the same held-out folds. Three findings. First, the skeleton graph helps only a little over the same network without it (macro recall 86.4% against 84.8%, while overall accuracy is 76.0% against 77.1%); differences of one or two points are smaller than the seed-to-seed variation we did not measure. Second, simpler models match or exceed the ST-GCN: the LSTM, BiLSTM and temporal CNN reach 89.5–89.7% macro recall, and an 86 k-parameter MLP on the window’s mean and standard deviation, which has no temporal modelling at all, is best (94.4%). Held-pose recognition is therefore largely a static-shape problem, and we do not claim that the graph is necessary for it. Third, every model shares one weakness: Warrior II (precision 0.24–0.30) and plank (0.10–0.40, 30 windows) are over-called because transition windows are labelled as poses, while transition/unknown itself has high precision (0.91–0.96). An ST-GCN fold takes about 290 s to train on a T4, against 6–15 s for the others. The absolute numbers differ from the production figures above (macro recall 86.4% against 78.0%, overall 76.0% against 82.2%) because this recipe uses fewer windows, fixed epochs and class weights, which trade overall accuracy for recall; the comparison within the table is what matters. We keep the ST-GCN as a second opinion and as the

basis for flow scores (smoothness, balance, momentum) that static statistics cannot provide, but the evidence for it today is parity, not superiority. *Does the network model movement?* Held-pose recognition is a static task, and the network is meant to capture how the body moves between poses, so we also tried to test movement. Our labels do not allow a named-transition benchmark: of the 568 consecutive hold pairs in the 24 videos, 205 are between different poses, and within 500 frames (70 transitions in 33 directional pairs) no pair occurs in three or more videos. We therefore built a rule-defined movement task: a window is a transition when its net angle change exceeds  $15^\circ/s$  and is named by the dominant pose of its first and last third, otherwise a hold or unrecognised, which gives 19 classes after dropping those with fewer than 100 windows or fewer than three videos (16 holds, generic transition, unrecognised, and one named transition, cobra pose to downward dog). The labels are defined by rules, so the task measures whether a model recovers rule-defined flow from the raw skeleton, not agreement with an annotator. The last column of Table IV shows the same ordering as for held poses: the window-statistics MLP is best (58.6% macro recall), the ST-GCN last (47.7%), and removing the skeleton graph changes nothing measurable (48.4%). A time-reversal test (every moving window shown forward and reversed, chance 50%) is at chance for every model (49.9–51.1%), so under this recipe no model, including the ST-GCN, is shown to use the direction of time; the test is uninformative, not evidence either way. We therefore make no claim that the network models movement; labelled transition data is the missing ingredient (Section VI).

*Does 3D matter?* The sequence models are the only models fed the depth coordinate  $z$  (the per-frame MLPs use angles computed with  $z=0$ ). Retraining the same models with  $z$  set to zero (Table III) lowers every model on both tasks: by 20 points (temporal CNN) and 35 points (MLP) on held poses, while the ST-GCN collapses from 88.7% to 28.6% macro recall, most windows then falling to the generic transition class; on the movement task the loss is 4 to 12 points. The depth coordinate is therefore the largest effect we measured, larger than the

**TABLE III:** Effect of the depth coordinate: macro recall of the same models trained with the full  $(x, y, z)$  skeleton and with  $z$  set to zero (bold: best in column). Held poses: nine classes on the cueT2 video folds; Movement: the 19-class rule-defined task. At most 3,000 training windows per class and one seed, so the full-skeleton held-pose values differ slightly from Table IV.

| Model                  | Held poses   |              | Movement     |              |
|------------------------|--------------|--------------|--------------|--------------|
|                        | $x, y, z$    | $z=0$        | $x, y, z$    | $z=0$        |
| MLP on window mean/std | <b>94.2%</b> | 59.3%        | <b>59.5%</b> | 46.3%        |
| Temporal CNN           | 90.5%        | <b>70.1%</b> | 56.2%        | <b>52.1%</b> |
| ST-GCN                 | 88.7%        | 28.6%        | 47.5%        | 35.9%        |

**TABLE IV:** Sequence models under one recipe on the held-out video folds (bold: best in column; for parameters, fewest). MR9: macro recall over the nine classes; MR8: over the eight poses; Pass: poses at recall and precision  $\geq 0.70$ ; Flow: macro recall over the 19 classes of the rule-defined movement task (24 videos, 3 folds by video).

| Model                     | Params        | Overall      | MR9          | MR8          | Pass     | Flow         |
|---------------------------|---------------|--------------|--------------|--------------|----------|--------------|
| LSTM (2 layers)           | 0.25 M        | 83.0%        | 89.5%        | 91.0%        | 4        | 53.1%        |
| BiLSTM + attention        | 0.50 M        | <b>87.1%</b> | 89.6%        | 90.1%        | <b>5</b> | 56.8%        |
| Temporal CNN              | 1.00 M        | 81.5%        | 89.7%        | 91.4%        | 4        | 55.6%        |
| MLP on window mean/std    | <b>0.09 M</b> | 85.3%        | <b>94.4%</b> | <b>96.4%</b> | <b>5</b> | <b>58.6%</b> |
| ST-GCN, no skeleton graph | 0.89 M        | 77.1%        | 84.8%        | 86.4%        | 4        | 48.4%        |
| ST-GCN (ours)             | 0.89 M        | 76.0%        | 86.4%        | 88.6%        | 4        | 47.7%        |

choice among sequence models, and the sequence input is how three-dimensional information enters the system. Caveats:  $z$  is MediaPipe’s monocular relative depth, not measured depth; the figures are for 24 videos and one seed; and the  $z$  a phone camera produces live may differ from that of the training videos.

### C. RQ3: hidden joints

We hid one limb with a grey box on 70 real yoga frames in four poses (141 cases) and recovered the hidden joints in three ways, with MediaPipe on the clean frame as the reference. Without recovery the median joint-angle error is  $17.9^\circ$ . Table V compares left–right mirroring with CLIFF [26]. Neither dominates: mirroring is better where the body is symmetric (arms, mountain, seated) and has the lowest median landmark error overall, CLIFF is better for legs, for both feet hidden and for the asymmetric Tree pose, and its median angle error is  $5.8^\circ$  against  $8.6^\circ$ . But CLIFF costs 164 ms per image on two CPU threads (18.7 ms on a T4 GPU) and, decisively, needs the image, which the app never sends. We therefore keep the abstain policy: a hidden joint is not scored, and the interface says which joints are not being checked. Caveats: 141 cases from 70 frames, one asymmetric pose, a synthetic occluder and MediaPipe, not motion capture, as the reference.

### D. RQ4: form score, personal profile and efficiency

*Form score.* No labelled good-versus-bad-form data exist for these poses, so we use a property test: on 648 held-out in-vocabulary photos, five trials each, one joint is broken by  $45^\circ$ . The gate’s form score is high on good form (0.81) and drops in 85.0% of trials, while the deviation head’s top-ranked joint is the broken one in only 12.4% of trials (chance 6.7%), which is why joint-level cues come from the angle bands. On the deployed endpoint with real MediaPipe angles from eight instructor clips the mean score is 0.94 while held, 0.25 with the left knee broken and 0.92 with the left elbow broken: the

**TABLE V:** Recovering hidden joints: median joint-angle error in degrees (bold: lower).

| Subset ( $n$ )          | Mirror     | CLIFF      |
|-------------------------|------------|------------|
| All (141)               | 8.6        | <b>5.8</b> |
| Symmetric stances (117) | <b>5.2</b> | 6.1        |
| Tree, asymmetric (24)   | 20.1       | <b>4.8</b> |
| Left leg hidden (47)    | 16.2       | <b>6.2</b> |
| Right arm hidden (47)   | <b>4.9</b> | 15.1       |
| Both feet hidden (47)   | 6.1        | <b>4.7</b> |

**TABLE VI:** Model size and measured latency.

| Item                                   | Measured                |
|----------------------------------------|-------------------------|
| Pose MLP (two models, cascade)         | 0.36 M parameters each  |
| ST-GCN flow check                      | 0.89 M parameters       |
| Server compute per frame, median (p95) | 7–9 ms (8–11 ms)        |
| Peak server memory, three networks     | 324 MB                  |
| Round trip, median (p95), $n=32$       | 967 ms (1,266 ms)       |
| Phone engine, 1 / 2 / 3 workers        | 7.3 / 19 / 24 results/s |
| Mesh recovery (CLIFF), 2 CPU threads   | 164 ms per image        |

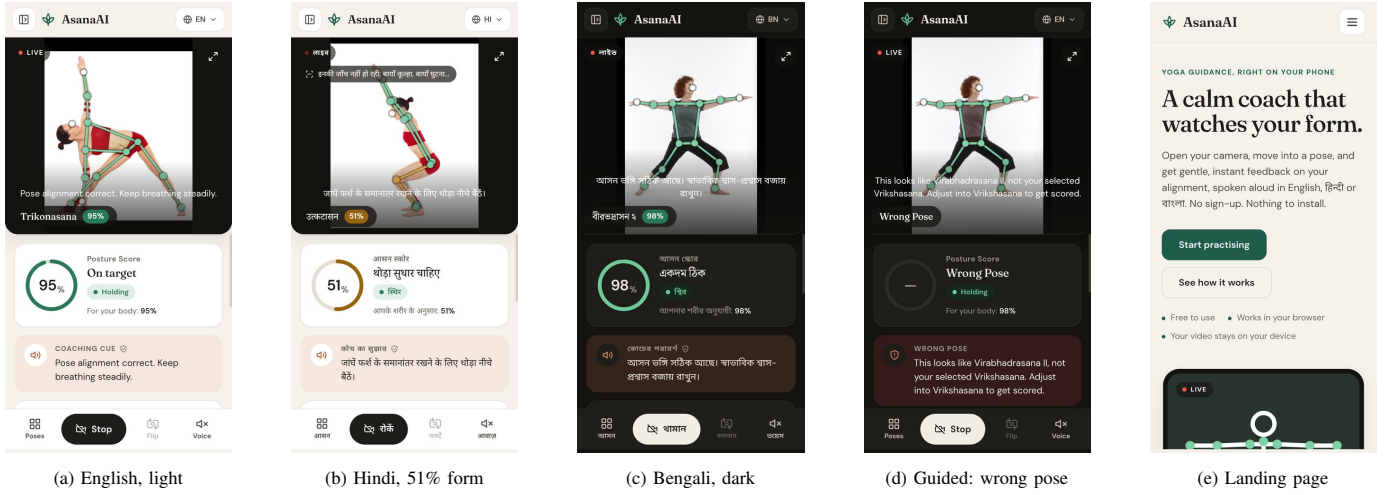
score collapses for leg errors and is nearly blind to arm errors (these clips are in the gate’s training data, so this checks the deployed path, not generalisation).

*Personal profile.* We tested the mechanism on the deployed server with a hand-built skeleton whose two hip deviations were  $56^\circ$  from their bands (universal score 0.924). Without a profile the response carries no personal score; a profile not containing the current angles leaves it at 0.924; one containing the left hip’s angle zeroes that deviation (0.962); both hips give 1.000; the universal score never changes. A profile spanning  $0^\circ$ – $180^\circ$  for every joint also gives 1.000, which shows how much a poor profile could forgive; the profile has not been validated with users.

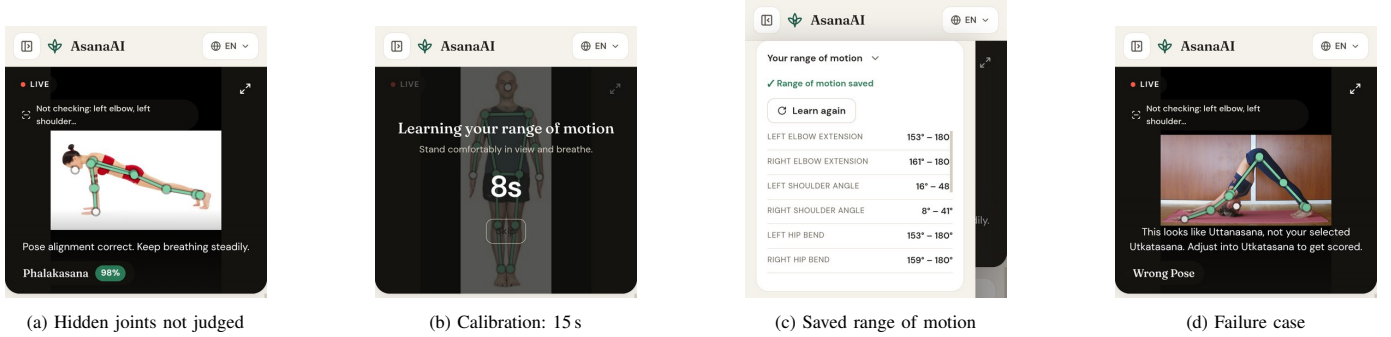
*Efficiency and hardware.* Table VI lists model size and latency. Server compute is 7–9 ms per frame (p95 8–11 ms) with 324 MB peak memory for all three networks, but the round trip from India is 0.97 s at the median because it is dominated by the network; the interface therefore draws the skeleton locally (about 22 updates/s on the phone) and refreshes score and cue about every 1.5 s, with the flow check and speech at about 10 s. On the phone the pose engine scales with the number of Web Workers (7.3, 19 and 24 results/s for one, two and three).

### E. End-to-end validation on a phone

We ran the real app on the phone through its CPU engine, with a Wikimedia photo standing in for the camera, and logged every answer of the live server; a photo counts if the most frequent of three answers is the expected pose, and a pose passes at 50% or more of its photos (Table VII). Eight of eleven poses pass; the six poses of the held-out benchmark that were tested all pass, while mountain, cobra and child’s pose do not. These are still photos with labels taken from file titles, and some may have been seen in training, so the test validates the on-phone path (engine, landmarks, server, label), not generalisation. Fig. 4 shows the interface in three languages, both themes and both modes, and Fig. 5 shows hidden-joint handling, the calibration screens and a failure case.



**Fig. 4:** The deployed interface on a POCO M7 Pro, a credited Wikimedia photograph standing in for the camera (no live person): (a) Trikonasana, 95%; (b) Utkatasana with the Hindi cue, 51% in amber and a chip naming the joints not checked; (c) Virabhadrasana II in Bengali; (d) guided mode with Vrikshasana selected while a Warrior II photo is shown: the score is withheld and the app names the pose it sees; (e) the public landing page. Photos by Kennguru (a, b; CC BY 3.0) and lululemon athletica (c, d; CC BY 2.0), via Wikimedia Commons.



**Fig. 5:** (a) Plank, guided mode: visible limbs are sage (inside their angle bands) and the chip “Not checking: left elbow, left shoulder” names joints the camera cannot see, which are drawn faint and neither scored nor coached. (b, c) The personal profile: a 15 s countdown and the saved per-joint ranges (the still photo yields narrow ranges). (d) A failure shown on purpose: a Downward Dog photo is named Uttanasana (a forward fold), after which guided mode answers “Wrong Pose”. Photos: Kennguru (a), Witold Fitz-Simon (b, c; CC BY-SA 2.5), Iveto (d); CC BY 3.0 unless noted, via Wikimedia Commons.

**TABLE VII:** On-phone photo test (photos correct / tested; a pose passes at 50% or more). Two photos had no person found.

| Pose         | Correct | Pose        | Correct | Pose         | Correct |
|--------------|---------|-------------|---------|--------------|---------|
| Tree         | 6/6     | Triangle    | 5/5     | Warrior II   | 3/3     |
| Downward dog | 4/4     | Seated easy | 4/5     | Plank        | 3/4     |
| Chair        | 3/5     | Corpse      | 2/4     | Child’s pose | 1/3     |
| Cobra        | 1/5     | Mountain    | 0/5     |              |         |

## VI. DISCUSSION

### A. How the approach answers the open problems

*G1, evaluation.* Every reported number comes from held-out photos or videos, with a false-alarm metric, a second wild photo set and five standard baselines on the same rows. The honest outcome is mixed: the cascade removes most false alarms of the rule engine (78.3% to 20.8%) and holds up on wild photos, but a random forest is stronger on photos from the training sources, which a random-split study would not have revealed. *G2, temporal behaviour.* The flow check recognises held poses and separates them from transitions on videos of unseen people (78.0% macro recall, three poses above the bar) and its input-rate requirement is measured. A

controlled comparison shows that simpler sequence models are as good for held-pose recognition and at least as good on a rule-defined movement task, and a time-reversal test is at chance for every model, so neither the graph nor the frame order is shown to be necessary, whereas the depth coordinate that only the sequence input carries is shown to matter (removing it costs every sequence model 4 to 60 points); our labels support only one named transition, so movement modelling is untested, and the flow check does not yet score smoothness, balance or momentum. *G3, personalisation.* A 15-second, vision-only range profile adapts the score without extra sensors; the mechanism behaves as designed but has not been validated with users. *G4, hidden joints.* The system declines to score what it cannot see, and the measured alternatives show that mirroring is better for symmetric stances and mesh recovery for legs and asymmetric poses, but the latter needs the image. *G5, generated advice.* Cues come from reviewed templates, an optional paraphrase may only rewrite them, a term screen can discard the rewrite, and repeated failures back off instead of pushing harder. *G6, deployment.* Only landmark numbers leave

the device, no extra hardware is needed, guidance is available in three languages, and a rule-engine port keeps a basic mode working offline.

### B. Cost-efficiency

Recent work often raises model complexity for marginal accuracy gains. Our design spends parameters sparingly: two 0.36M-parameter MLPs and one 0.89M-parameter ST-GCN run in 324MB on a 2-vCPU host at 7–9 ms per frame. The measurements show where complexity pays and where it does not. The second, small MLP (the gate) costs about one more forward pass and cuts false alarms from 81.5% (namer alone) to 20.8%; residual blocks add nothing over a plain MLP on public photos, and an 86k-parameter MLP on window statistics matches or exceeds the 0.89M-parameter ST-GCN at held-pose recognition while training in 6 s instead of 290 s per fold; and mesh recovery improves the median angle error by 2.8° at 164 ms per image on a CPU and needs the camera image, so we do not use it.

### C. Limitations

- *No live-person accuracy study.* All end-to-end tests use photographs or recorded video; a study with practitioners and instructors is needed.
- *The form score is not validated against human judgement.* It is checked only by a perturbation test, it is nearly blind to arm errors (0.94 to 0.92 when an elbow is broken), and the learned deviation head is close to chance (12.4% against 6.7%); joint-level cues rest on hand-set angle bands, and Tree and Lunge have none.
- *Weak poses.* Mountain, cobra and child’s pose fail the phone photo test; the flow check reliably handles three poses and is matched by far simpler sequence models; our labels name only one transition (cobra pose to downward dog) with support in three or more videos, so movement modelling is untested; named transitions and momentum scores are not implemented.
- *Small data.* 24 videos and public photo sets; the held-out public photos share their sources with the training photos; the wild set has 103 photos; baselines use one seed.
- *Occlusion.* Pose naming can still be influenced by a hallucinated hidden joint; the mesh-recovery study is small (141 cases, one asymmetric pose, a synthetic occluder).
- *Personal profile and safety.* The profile is not user-tested and a poor stance could forgive real errors; a term screen cannot certify open-ended text, the Hindi and Bengali templates and term lists were checked by one native speaker of both languages on the team and have not had an independent review by instructors.
- *Latency and hardware.* The round trip is about 1 s from India because it is network-dominated; one phone model was tested. The system is not a medical device and no injury outcome was studied.

### D. Future work

(i) A live user study in which instructors rate form, producing the labelled good/bad-form data the form head lacks, in the

manner of recent isometric-pose benchmarks [23]; (ii) a tree–network ensemble for naming, since the two families win on different sources; (iii) on-device inference of the 0.36M-parameter MLPs, which would remove the round trip; (iv) hand-set bands for Tree and Lunge; (v) flow scores (smoothness, balance, momentum) and named transitions, the tasks on which a temporal model must prove its value over static statistics; (vi) retraining the form head with arm perturbations; (vii) viewpoint-robust features and a larger pose vocabulary; (viii) opt-in server-side mesh recovery with explicit image consent.

## VII. CONCLUSION

AsanaAI shows that a yoga coach can keep the camera image on the device, abstain when evidence is missing and still run on commodity hardware, provided it is evaluated on data it never trained on. A name-and-veto cascade of small residual MLPs names poses on 78.9% of held-out photos with 20.8% false alarms, a skeleton-graph flow check reaches 78.0% macro recall on held-out videos although simpler sequence models are as good on held poses and on a rule-defined movement task, and the system degrades gracefully through three pose engines and an offline mode. The comparison with standard methods shows that no single classifier wins on every photo source, and we report what the system cannot yet do.

*Availability.* App: <https://yoga-posture-correction-system.vercel.app>; code: <https://github.com/Anamitra-Sarkar/yoga-posture-correction-system>; training data and code: Hugging Face dataset Arko007/Yoga-1M (transcripts withheld because they are copyrighted speech).

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